

SOUTHLAND FIELD, LA, USA

WMO: 722248

Lat: 30.131N

Lon: 93.376W

Elev: 10

StdP: 14.69

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 00359

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 31.6	(c) 34.1	(d) 19.2	(e) 14.4	(f) 37.9	(g) 24.7	(h) 18.8	(i) 44.3	(j) 23.1	(k) 58.5	(l) 20.4	(m) 60.5	(n) 5.1	(o) 340	(p) 0.398

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.5	96.6	79.0	94.7	78.4	92.6	78.2	83.7	89.8	82.0	87.7	81.1	86.8	7.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.0	166.6	85.8	81.1	161.6	84.7	79.4	152.5	83.2	47.5	89.6	45.7	89.5	44.5	85.1	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.4	(b) 18.2	(c) 16.4	(e) 26.1	(f) 98.9	(g) 4.2	(h) 2.6	(i) 23.1	(j) 100.8	(k) 20.6	(l) 102.3	(m) 18.2	(n) 103.8	(o) 15.2	(p) 105.7		
(4) 20.4	(5) 18.2	(6) 16.4	(7) 23.9	(8) 83.4	(9) 3.8	(10) 2.0	(11) 21.2	(12) 84.8	(13) 18.9	(14) 86.0	(15) 16.8	(16) 87.1	(17) 14.0	(18) 88.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	70.0	52.7	57.2	63.4	69.2	76.8	82.7	83.4	84.2	80.6	71.7	61.6	56.3
	DBStd	13.07	9.56	9.83	8.15	6.45	4.86	3.00	3.16	2.67	3.75	7.46	9.05	9.19
	HDD50	189	83	42	9	1	0	0	0	0	0	0	14	40
	HDD65	1310	395	248	129	35	4	0	0	0	0	26	171	302
	CDD50	7504	166	243	425	577	829	982	1036	1061	917	673	361	234
	CDD65	3151	14	30	80	161	368	532	571	596	467	233	68	31
	CDH74	31301	23	66	308	989	3213	6025	6653	7015	4518	2062	334	95
	CDH80	12667	0	1	14	127	991	2689	3023	3310	1817	654	40	1
(14) Wind	WSAvg	6.6	7.2	8.2	8.1	8.8	7.4	5.9	4.7	4.6	4.7	5.9	6.5	7.1
(15) Precipitation	PrecAvg	59.7	6.1	3.9	3.9	4.3	4.6	6.1	6.5	4.9	5.3	5.1	4.6	5.0
	PrecMax	83.0	14.3	8.3	8.7	8.4	12.8	12.5	12.5	9.9	12.2	19.2	12.4	10.8
	PrecMin	37.9	1.7	0.7	0.6	0.5	0.1	1.0	2.9	0.9	1.0	1.3	0.8	2.4
	PrecStd	12.0	3.2	2.3	2.0	2.3	3.1	3.2	2.4	2.3	3.0	4.7	2.6	2.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.9	79.2	81.8	86.0	91.4	99.0	97.1	99.3	95.3	91.0	84.1	79.3
		MCWB	67.8	69.8	68.8	69.8	73.9	77.1	78.3	80.8	76.0	75.2	73.6	71.7
	2%	DB	73.4	76.3	79.7	82.7	89.5	95.0	95.0	97.1	92.8	88.7	81.0	77.0
		MCWB	66.2	68.8	70.0	71.3	74.1	78.4	78.3	80.2	76.8	74.9	71.5	70.8
	5%	DB	71.2	73.3	77.6	81.4	87.5	92.1	93.1	95.1	90.9	86.6	78.2	73.7
		MCWB	65.8	68.2	68.4	71.1	74.8	77.9	78.5	79.3	76.9	74.0	69.3	69.6
	10%	DB	68.1	71.4	75.2	79.4	85.7	90.7	91.3	92.6	89.4	83.9	75.0	71.3
		MCWB	63.9	66.8	67.8	70.2	75.0	78.0	78.6	79.0	76.4	72.9	68.7	68.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.4	72.5	73.4	77.0	80.4	84.8	84.6	86.3	81.6	81.1	76.4	74.6
		MCDB	74.2	76.4	78.0	80.5	82.9	87.9	88.1	92.3	87.1	84.8	79.9	77.1
	2%	WB	70.0	71.3	72.4	75.2	78.5	82.0	82.3	84.2	80.3	79.1	74.3	72.8
		MCDB	72.0	74.6	76.5	79.9	82.9	87.0	87.3	91.5	86.3	83.7	77.7	75.4
	5%	WB	68.0	70.1	71.3	74.2	77.5	81.0	81.2	82.5	79.3	77.7	72.7	71.0
		MCDB	69.5	72.3	75.0	78.7	83.1	86.5	86.7	89.2	85.4	82.7	76.1	73.2
	10%	WB	64.6	68.4	70.2	73.1	76.7	80.0	80.4	81.3	78.4	76.2	70.4	69.1
		MCDB	66.8	70.1	73.5	77.5	82.5	86.2	86.4	87.5	84.4	81.3	74.0	70.7
(35) Mean Daily Temperature Range	5% DB	MDBR	19.1	17.2	18.0	17.7	16.1	15.6	15.9	16.5	16.6	20.4	19.5	17.6
		MCDBR	19.8	17.0	16.3	17.6	17.7	18.0	17.9	20.1	18.0	19.0	18.9	18.3
		MCWBR	13.7	11.2	9.2	8.5	6.5	5.7	5.6	6.3	5.8	7.9	10.4	12.4
	5% WB	MCDBR	15.8	13.8	14.0	14.3	13.7	15.4	15.4	17.2	15.1	15.8	15.8	16.7
		MCWBR	13.1	10.3	9.9	8.8	6.6	6.6	6.4	7.5	5.9	8.7	10.9	13.2
(40) Clear-Sky Solar Irradiance	taub		0.341	0.358	0.376	0.430	0.467	0.502	0.500	0.487	0.458	0.383	0.368	0.351
	taud		2.515	2.478	2.421	2.267	2.198	2.136	2.183	2.223	2.270	2.463	2.447	2.487
	Ebn at Noon		283	288	289	275	264	253	253	256	260	275	270	272
	Edh at Noon		28	32	36	43	47	49	47	45	41	32	30	28
(44) All-Sky Solar Radiation	RadAvg		908	1033	1435	1737	1940	1949	1879	1780	1585	1393	1050	805
	RadStd		89	170	130	130	129	149	147	161	158	176	101	89

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (31 neighbors)	+0.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+189	+144

Nomenclature: See separate page